

How NTT Works - Confirmed Understanding

This document is the algorithm structure as worked out and confirmed step by step:
256 coefficients in the polynomial, 128 zeta numbers in the list, 7 rounds total.

The one formula, used every single time

Take a pair of numbers (a, b) and the round's zeta:

$$\text{new pair} = (a + \text{zeta} \times b, a - \text{zeta} \times b) \bmod q$$

How the zeta list gets used

There is ONE shared list of 128 zetas. There is ONE counter that starts at 1 and goes up by 1 every time ANY group, in any round, starts. Every group just grabs the next unused zeta from this single list - nothing is calculated, just read off in order.

The 7 rounds

Round	Groups	Zetas used	Pairing distance
1	1	1 (index 1)	128 apart
2	2	2 (indices 2-3)	64 apart
3	4	4 (indices 4-7)	32 apart
4	8	8 (indices 8-15)	16 apart
5	16	16 (indices 16-31)	8 apart
6	32	32 (indices 32-63)	4 apart
7	64	64 (indices 64-127)	2 apart

$1 + 2 + 4 + 8 + 16 + 32 + 64 = 127$ - every single zeta in the list gets used exactly once.

A real worked example - position 0, all 7 rounds

Round 1: (2619, 2044), zeta=1729 -> becomes 1297
Round 2: (1297, 1532), zeta=2580 -> becomes 2334
Round 3: (2334, 366), zeta=2642 -> becomes 567
Round 4: (567, 1293), zeta=1062 -> becomes 2185
Round 5: (2185, 3293), zeta=296 -> becomes 1516
Round 6: (1516, 3007), zeta=289 -> becomes 1670
Round 7: (1670, 1557), zeta=17 -> becomes 1507

Important: this is the WHOLE NTT transform, but only ONE part of a multiplication

Everything above (all 7 rounds) is the complete NTT(f) function on its own - not a partial piece. To actually multiply two polynomials, this same 7-round process runs TWICE (once on f, once on g), then a separate multiply step combines them (using a different table, zetas_mult, pair by pair), then a reversed version of this same 7-round process converts the result back into a normal polynomial.